

Weekly Progress Report

Name: Keshavarapu Suresh

Domain: Java Full Stack Development

Date of submission: 15-07-2026

Week Ending: 05

I. Overview:

This week, I developed an **Advanced Music Player with Playlist Management System** using **Java, JDBC, and MySQL Database** in Eclipse IDE. The application is a menu-driven console application that allows users to manage a music collection, create playlists, add and remove songs from playlists, view playlists, and simulate music playback. The project uses JDBC for database connectivity to store and retrieve song and playlist information.

II. Achievements

1. Advanced Music Player Development

Project Name: Music Player with Playlist (Java + JDBC + Database)

- Created a Java Project in Eclipse IDE.
- Configured Java Build Path.
- Added MySQL JDBC Connector.
- Created MySQL Database.
- Created Songs table.
- Created Playlists table.
- Created Playlist_Songs table.
- Established JDBC Connection.
- Developed menu-driven application.
- Displayed available songs from the database.
- Implemented playlist creation.
- Added songs to playlists.

- Displayed songs inside playlists.
- Removed songs from playlists.
- Implemented Play Song functionality.
- Implemented Pause Song functionality.
- Implemented Stop Song functionality.
- Implemented Skip Song functionality.
- Tested database operations successfully.
- Implemented file handling using expenses.txt.
- Successfully tested and executed the application.

2. Project Files Created

- Main.java
- MusicPlayer.java
- DBConnection.java
- Song.java
- Playlist.java
- mysql-connector-j-9.7.0.jar
- musicplayer.sql

3. Database Tables Created

songs

Stores song details such as:

- Song ID
- Song Title
- Artist
- Album
- Duration

playlists

Stores playlist information:

- Playlist ID

- Playlist Name

playlist_songs

Stores the relationship between songs and playlists.

4. Features Implemented

Music Library

- View available songs
- Store songs in MySQL Database
- Retrieve songs using JDBC

Playlist Management

- Create Playlist
- Add Songs to Playlist
- View Playlist
- Remove Songs from Playlist

Music Controls

- Play Song
- Pause Song
- Stop Song
- Skip Song

Database Operations

- INSERT
- SELECT
- DELETE
- SQL JOIN Queries

5. Skills Learned

- Core Java
- Object-Oriented Programming (OOP)

- JDBC Connectivity
- MySQL Database
- SQL Queries
- PreparedStatement
- ResultSet
- Exception Handling
- Menu-driven Programming
- CRUD Operations
- Database Design

III. Challenges Faced

Challenge 1

Connecting Java with MySQL Database.

Solution

Downloaded the MySQL JDBC Driver, added it to the Eclipse Build Path, and configured the database connection using JDBC.

Challenge 2

Managing playlists using multiple database tables.

Solution

Created three related tables (songs, playlists, and playlist_songs) and used SQL JOIN queries to retrieve playlist data.

Challenge 3

Displaying playlist songs.

Solution

Implemented SQL JOIN operations with PreparedStatement and ResultSet to fetch playlist contents.

Challenge 4

Implementing playback controls.

Solution

Developed Play, Pause, Stop, and Skip methods to simulate music player controls within the console application.

IV. Learning Resources

- Oracle Java Documentation
- MySQL Documentation
- JDBC Documentation
- Eclipse IDE Tutorials
- Java Collections Framework
- Java Exception Handling
- SQL Tutorials

V. Next Week's Goals

- Implement real MP3 playback.
- Add Search Songs feature.
- Add Update Song Details feature.
- Improve exception handling.
- Develop a Java Swing GUI.
- Enhance the user interface.
- Optimize database queries.

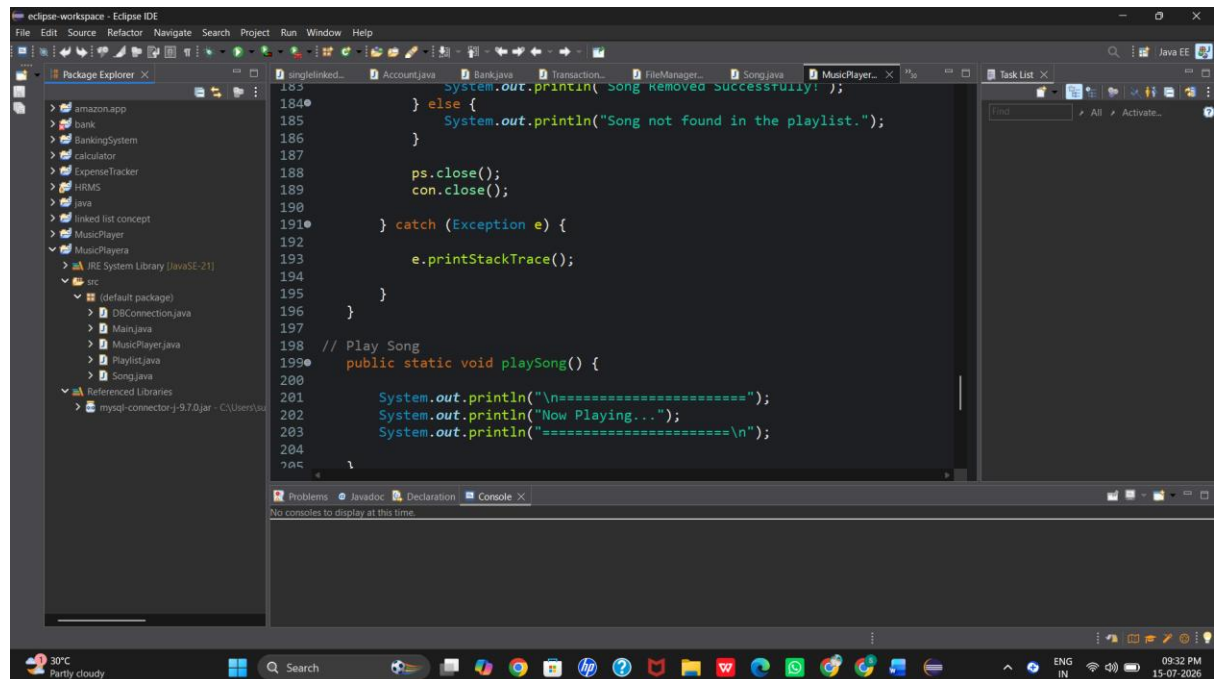
VI. Additional Comments

This advanced project provided hands-on experience in developing a database-driven Java application using JDBC and MySQL. I improved my understanding of database connectivity, SQL queries, object-oriented

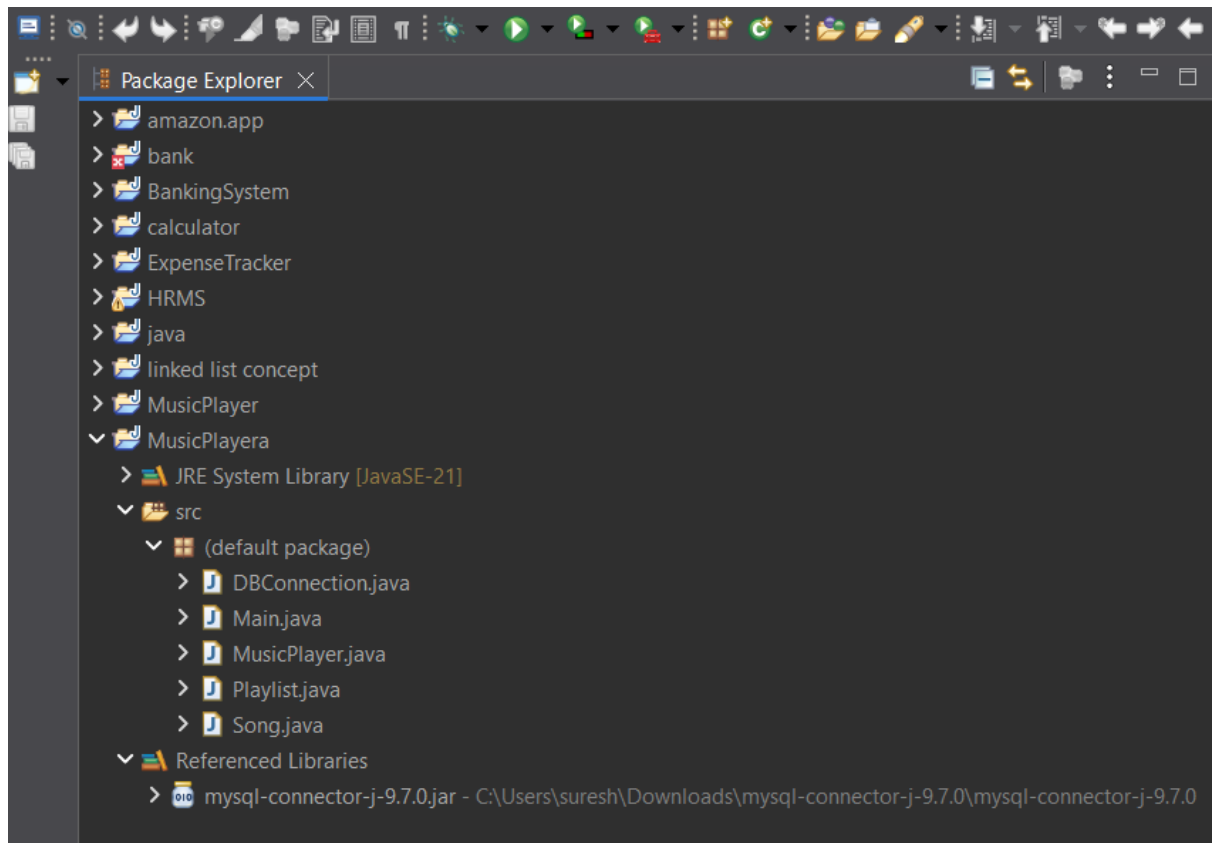
programming, and menu-driven application development. Successfully implementing playlist management and music control features strengthened my practical Java development skills.

VII. Screenshots Report

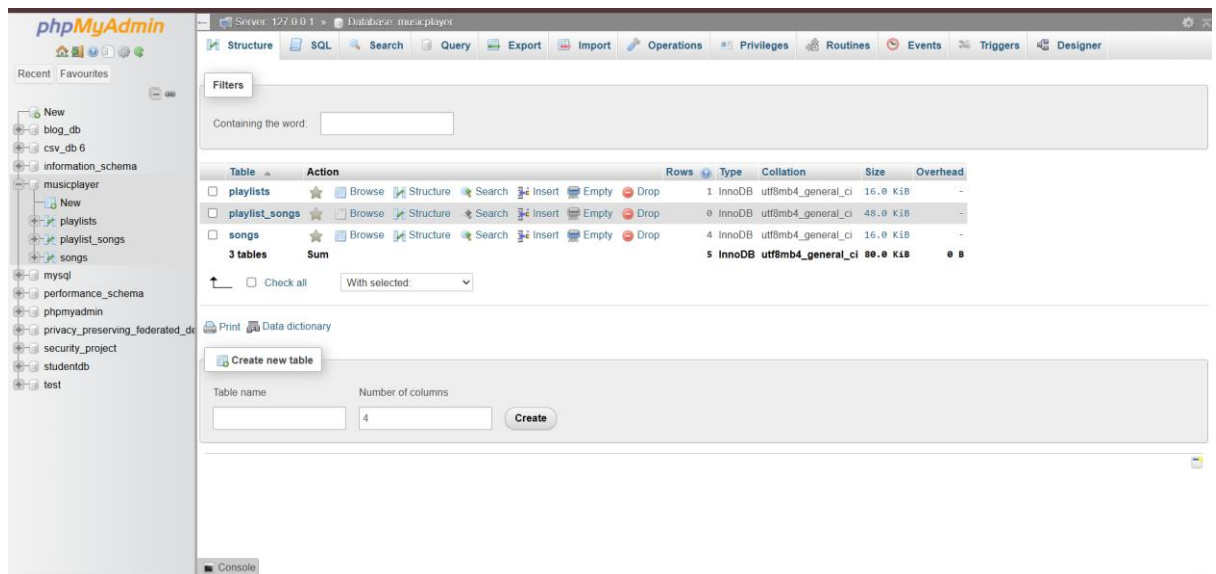
1. Eclipse Project Structure



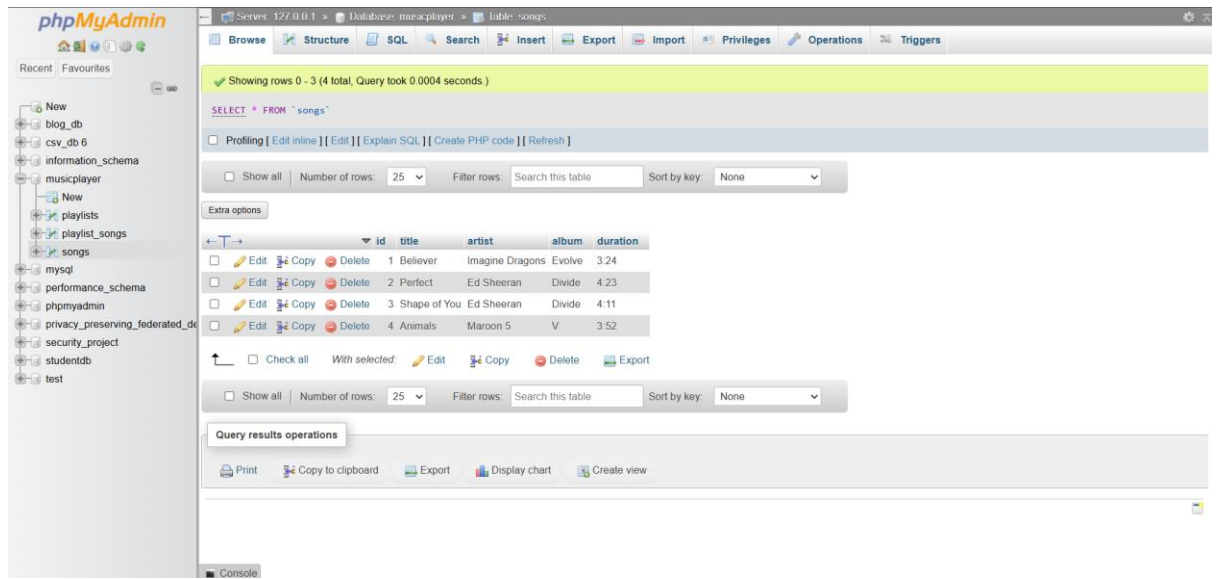
2. Java Package Explorer



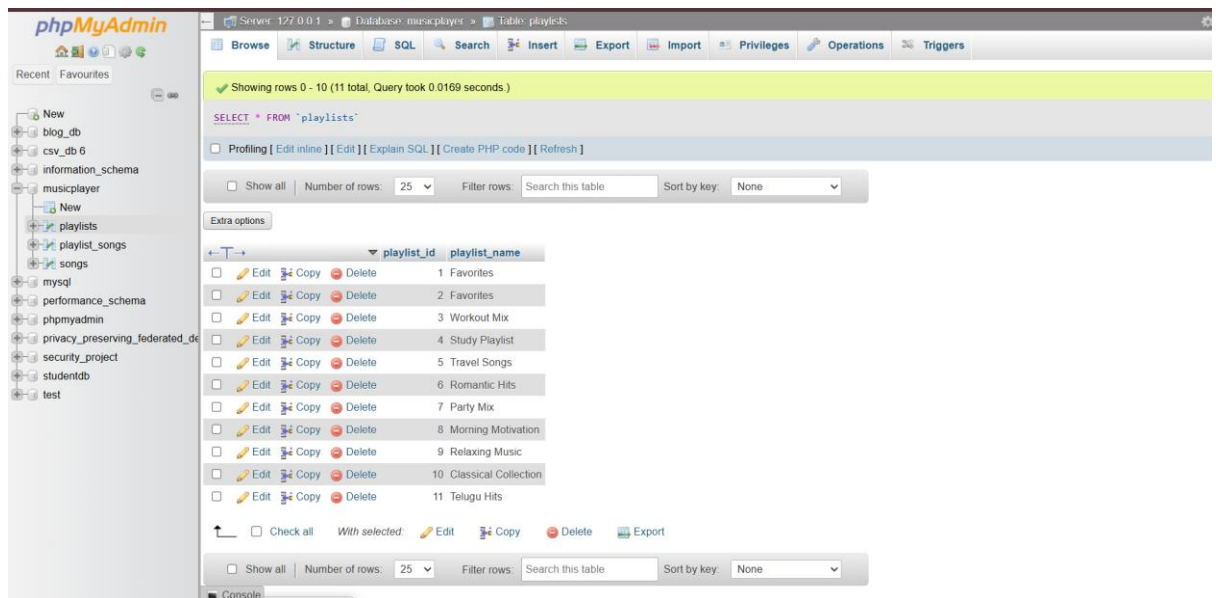
3. MySQL Database (musicplayer)



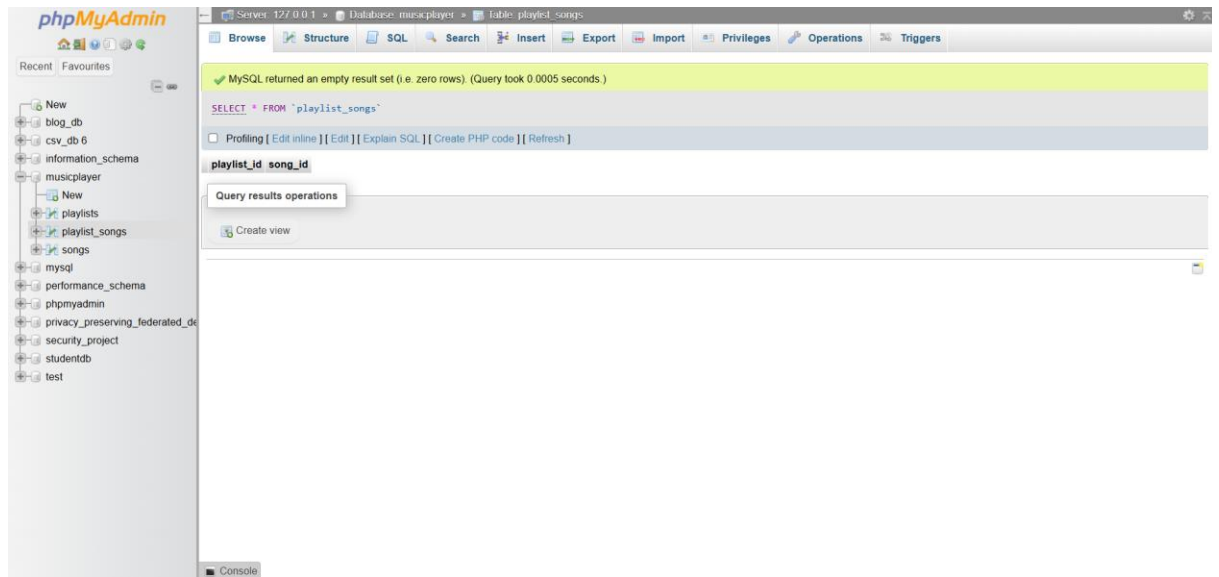
4. Songs Table



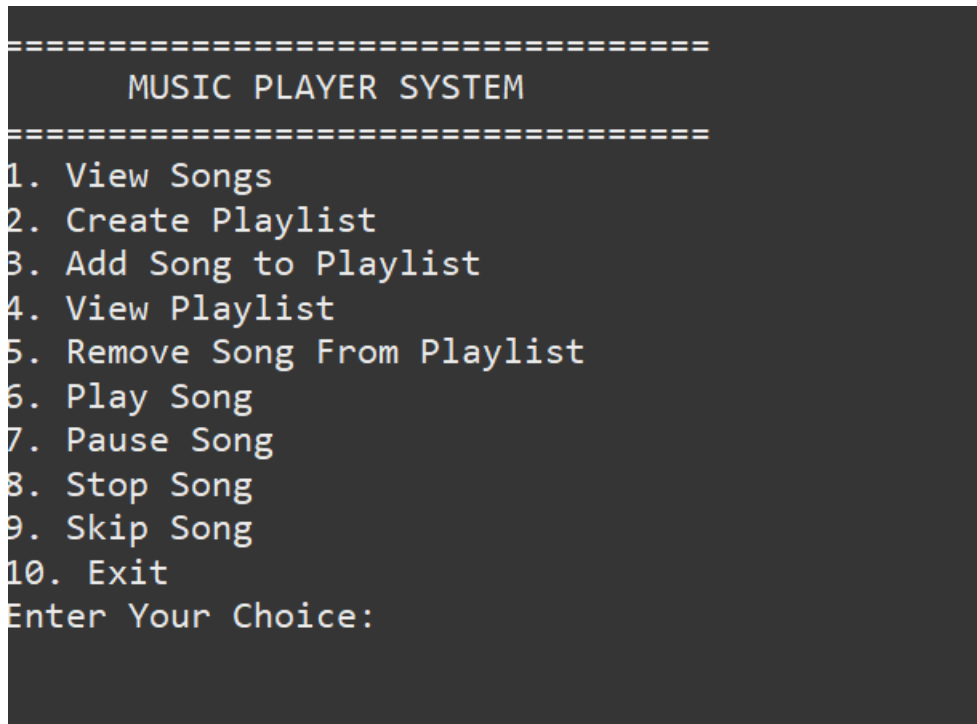
5. Playlists Table



6. Playlist_Songs Table



7. Main Menu Output



8. View Songs Output

```

=====
          MUSIC PLAYER SYSTEM
=====
1. View Songs
2. Create Playlist
3. Add Song to Playlist
4. View Playlist
5. Remove Song From Playlist
6. Play Song
7. Pause Song
8. Stop Song
9. Skip Song
10. Exit
Enter Your Choice: 1
|
===== SONG LIST =====
1 | Believer | Imagine Dragons | Evolve | 3:24
2 | Perfect | Ed Sheeran | Divide | 4:23
3 | Shape of You | Ed Sheeran | Divide | 4:11
4 | Animals | Maroon 5 | V | 3:52

```

9. Create Playlist Output

```

=====
          MUSIC PLAYER SYSTEM
=====
1. View Songs
2. Create Playlist
3. Add Song to Playlist
4. View Playlist
5. Remove Song From Playlist
6. Play Song
7. Pause Song
8. Stop Song
9. Skip Song
10. Exit
Enter Your Choice: 2
Enter Playlist Name : Favorites
Playlist Created Successfully.

```

10. Add Song to Playlist Output

```

=====
      MUSIC PLAYER SYSTEM
=====
1. View Songs
2. Create Playlist
3. Add Song to Playlist
4. View Playlist
5. Remove Song From Playlist
6. Play Song
7. Pause Song
8. Stop Song
9. Skip Song
10. Exit
Enter Your Choice: 3
Enter Playlist ID : 1
Enter Song ID : 2
Song Added Successfully.

```

11. View Playlist Output

```

=====
      MUSIC PLAYER SYSTEM
=====
1. View Songs
2. Create Playlist
3. Add Song to Playlist
4. View Playlist
5. Remove Song From Playlist
6. Play Song
7. Pause Song
8. Stop Song
9. Skip Song
10. Exit
Enter Your Choice: 4
Enter Playlist ID: 1

===== PLAYLIST SONGS =====
2 | Perfect | Ed Sheeran | Divide | 4:23
2 | Perfect | Ed Sheeran | Divide | 4:23

```

12. Remove Song Output

```
=====
      MUSIC PLAYER SYSTEM
=====
1. View Songs
2. Create Playlist
3. Add Song to Playlist
4. View Playlist
5. Remove Song From Playlist
6. Play Song
7. Pause Song
8. Stop Song
9. Skip Song
10. Exit
Enter Your Choice: 5
Enter Playlist ID: 1
Enter Song ID: 2
Song Removed Successfully!
```

13. Play Song Output

```
=====
      MUSIC PLAYER SYSTEM
=====
1. View Songs
2. Create Playlist
3. Add Song to Playlist
4. View Playlist
5. Remove Song From Playlist
6. Play Song
7. Pause Song
8. Stop Song
9. Skip Song
10. Exit
Enter Your Choice: 6

=====
Now Playing...
=====
```

14. Pause Song Output

```
=====
      MUSIC PLAYER SYSTEM
=====
1. View Songs
2. Create Playlist
3. Add Song to Playlist
4. View Playlist
5. Remove Song From Playlist
6. Play Song
7. Pause Song
8. Stop Song
9. Skip Song
10. Exit
Enter Your Choice: 7

=====
Song Paused
=====
```

15. Stop Song Output

```
=====
      MUSIC PLAYER SYSTEM
=====
1. View Songs
2. Create Playlist
3. Add Song to Playlist
4. View Playlist
5. Remove Song From Playlist
6. Play Song
7. Pause Song
8. Stop Song
9. Skip Song
10. Exit
Enter Your Choice: 8

=====
Song Stopped
=====
```

16. Skip Song Output

```
=====
      MUSIC PLAYER SYSTEM
=====
1. View Songs
2. Create Playlist
3. Add Song to Playlist
4. View Playlist
5. Remove Song From Playlist
6. Play Song
7. Pause Song
8. Stop Song
9. Skip Song
10. Exit
Enter Your Choice: 9
|
=====
Playing Next Song...
=====
```

17. Project Execution Success

```
=====
      MUSIC PLAYER SYSTEM
=====
1. View Songs
2. Create Playlist
3. Add Song to Playlist
4. View Playlist
5. Remove Song From Playlist
6. Play Song
7. Pause Song
8. Stop Song
9. Skip Song
10. Exit
Enter Your Choice: 10
|
Thank You for Using Music Player!
```